Case Study: Forecasting Bitcoin Volatility

GARCH, Realized Variance and Out-of-Sample Risk Forecasting

Predictive Statistics in Finance

The decision

You are part of the **risk analytics team of a digital-asset investment firm**. The portfolio manager must decide how much Bitcoin risk the portfolio should carry tomorrow. A useful volatility forecast can inform position sizing, risk limits, margin decisions and, later in the course, measures such as Value at Risk.

The objective is not to find the model that describes the historical sample most elegantly. The firm needs a model that produces **useful forecasts of future risk**. Your central question is:

How predictable is tomorrow's Bitcoin volatility, and what forecasting approach should the firm actually use?

A good answer need not identify a model that forecasts volatility perfectly. Your job is to determine what the models learn, how well that information survives out of sample, and what the results can realistically be used for.

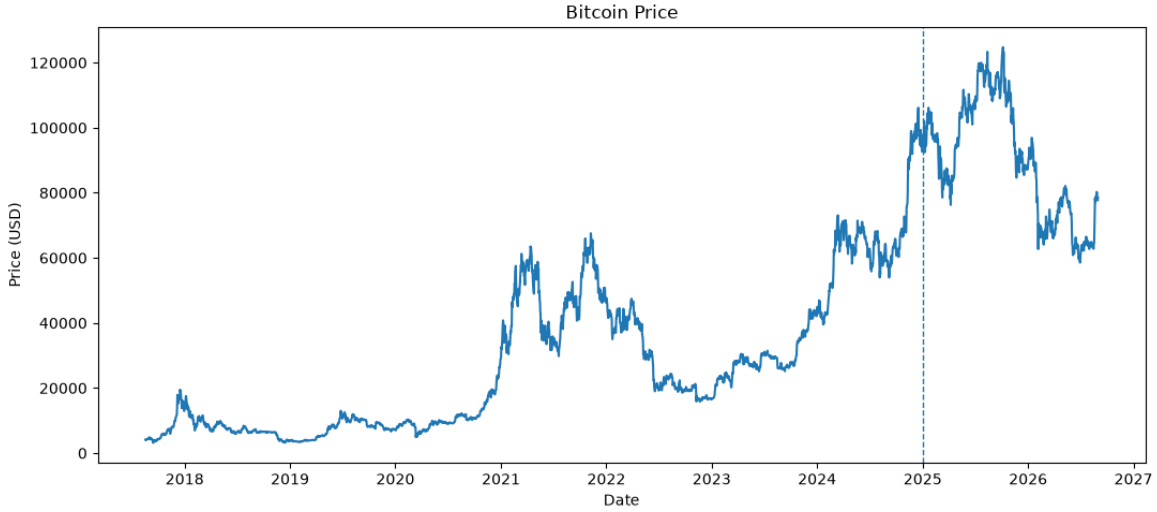


Figure 1: Bitcoin price and beginning of the out-of-sample evaluation period. The dashed vertical line marks 1 January 2025.

Before estimating any model, look at Figure 1. Bitcoin has experienced very different market episodes over its observed history. Keep this visual context in mind throughout the case: *what features of Bitcoin's history might make current risk difficult to forecast using distant observations?*

1 Data and timing

You are provided with curated Bitcoin data so that the exercise focuses on modelling rather than data acquisition. The daily price sample runs from **17 August 2017 to 31 August 2026**. Daily

continuously compounded percentage returns are

$$r_t = 100[\log(P_t) - \log(P_{t-1})].$$

The first return is therefore dated 18 August 2017.

The main forecast evaluation period is **1 January 2025–31 August 2026**. All model choices that use this period must respect the information available at each forecast date. For a forecast of day $t+1$, do not use observations from $t+1$ or later.

You are also provided with a daily **realized variance** (RV) series constructed from **24 intra-day hourly Bitcoin returns**. For each calendar day t, let $r_{t,j}$ denote the percentage log return during hour j, for $j = 1, \ldots, 24$. Daily realized variance is

$$RV_t = \sum_{j=1}^{24} r_{t,j}^2.$$

Thus, instead of using only one squared daily return to learn about the variance realized during a day, RV uses the information contained in the 24 hourly price movements. Think of the two main objects carefully:

$$\sigma_{t+1|t}^2 = \mathrm{Var}(r_{t+1} \mid \mathcal{F}_t), \qquad RV_{t+1} = \sum_{j=1}^{24} r_{t+1,j}^2.$$

The first is the **latent conditional variance** that a GARCH model attempts to forecast. The second is an **ex-post proxy** for the variance realized during the following day. Realized variance is much more informative than a single squared daily return, but it is **not the "true" latent conditional variance**. It remains an imperfect proxy for the object the model seeks to forecast. In this case, we therefore use RV as our **best available empirical proxy** for evaluating volatility forecasts, while keeping this measurement distinction in mind.

2 Part I — Is there volatility to forecast?

Task 1 — Explore Bitcoin returns

Using the training sample ending 31 December 2024, produce a compact exploratory analysis. Include a plot of daily returns, descriptive statistics, and evidence on dependence in returns and squared returns. You may use ACFs and/or an ARCH-LM test.

Discuss:

(a) Are daily Bitcoin returns themselves strongly predictable from their own recent history?

(b) Is there evidence of **volatility clustering**?

(c) What does an ARCH-LM rejection establish, and what does it *not* establish?

Do not include output simply because Python produces it. Every table or graph should help answer the forecasting question.

3 Part II — ARCH and GARCH

Consider

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t.$$

An ARCH(1) model specifies

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2,$$

whereas GARCH(1,1) specifies

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

Task 2 — Why does GARCH help?

Estimate ARCH(1) and GARCH(1,1) under conditionally Normal innovations using the training sample. Compare their fit using log-likelihood and information criteria, but focus your discussion on the economic meaning of the variance dynamics.

For GARCH(1,1), interpret $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\alpha} + \hat{\beta}$. In particular, answer: **how strongly does estimated risk react to a new return shock, and how persistent is that increase in risk?** If $0 < \hat{\alpha} + \hat{\beta} < 1$, you may also compute the volatility-shock half-life.

Report the statistical significance of the main variance parameters. Explain why a statistically significant volatility relationship is not yet evidence that the model will forecast future volatility accurately.

4 Part III — Does the conditional distribution matter?

Task 3 — Normal versus Student-t GARCH

Estimate GARCH(1,1) using both conditionally Normal and standardized Student-t innovations. Compare log-likelihood, AIC and BIC. For the Student-t specification, report and interpret the estimated degrees-of-freedom parameter ν.

Discuss whether allowing for heavy tails materially changes the statistical description of Bitcoin returns. Keep two modelling choices separate: the **conditional variance recursion** and the **conditional distribution of the innovation**.

5 Part IV — Forecast a latent variable

The firm's actual objective is prediction. Using an **expanding-window pseudo-out-of-sample exercise**, produce one-day-ahead Student-t GARCH(1,1) variance forecasts for every date from 1 January 2025 through 31 August 2026.

For each forecast date $t + 1$:

1. estimate the model using only returns available through t;
2. produce $\widehat{\sigma}^2_{t+1|t}$;
3. compare it with RV_{t+1} after that day has occurred.

Task 4 — Evaluate the GARCH forecasts

Plot the GARCH variance forecasts together with realized variance. Evaluate the forecasts using

$$MSE = \frac{1}{T}\sum_{t=1}^{T}(RV_t - \widehat{h}_t)^2$$

and

$$QLIKE = \frac{1}{T}\sum_{t=1}^{T}\left[\log(\widehat{h}_t) + \frac{RV_t}{\widehat{h}_t}\right].$$

Also report

$$\mathrm{Corr}(\widehat{h}_t, RV_t).$$

Explain what each statistic tells you. In particular, why can a model have a relatively high forecast correlation and still have poor MSE or QLIKE? Compare the average forecast variance with average realized variance and discuss calibration.

Then address a conceptual question: **if GARCH forecasts high variance for tomorrow and tomorrow happens to be quiet, does that single observation prove that the variance forecast was wrong?** Relate your answer to the fact that conditional variance is latent.

6 Part V — A simple realized-variance benchmark

A colleague proposes forecasting the observable volatility proxy directly:

$$RV_{t+1} = a + bRV_t + u_{t+1}.$$

Estimate this model using the training information and generate one-day-ahead forecasts over **exactly the same 2025–2026 OOS dates** as GARCH. Call this the **RV-OLS benchmark**.

Task 5 — Significance versus forecasting performance

Report the estimated b, its statistical significance, and the in-sample R^2. Interpret b as persistence in realized variance. Then compare RV-OLS and GARCH-t using the same OOS MSE, QLIKE and forecast correlation. Also compare the **dynamics of their forecasts**. The RV-OLS benchmark responds to the latest realized variance through the simple linear rule $a + bRV_t$, whereas GARCH updates its variance forecast through both the latest return innovation and the previous conditional variance. Examine the forecast paths and discuss whether GARCH appears **more dynamically flexible** around changes in market volatility, and whether that extra flexibility translates into better OOS forecast performance.

Your discussion must distinguish these two questions:

Does today's realized variance contain statistically significant information about tomorrow's realized variance?

Does the resulting model forecast new observations accurately enough to be useful?

Explain why a highly significant coefficient does not guarantee excellent OOS forecasting performance.

7 Part VI — Does distant Bitcoin history still help?

The expanding-window procedure retains all available history. But Bitcoin's market structure and volatility environment may evolve. Investigate a second forecasting rule that uses only the **most recent two years** at each forecast origin.

For both GARCH-t and RV-OLS, repeat the 2025–2026 forecast exercise using a **two-year rolling estimation window**. The OOS evaluation dates must remain unchanged. Only the estimation information set changes. For the **first OOS forecast, 1 January 2025**, estimate each model using observations from **1 January 2023 through 31 December 2024**. For the forecast of 2 January 2025, move the window forward one day and use **2 January 2023 through 1 January 2025**; continue in this way throughout the OOS period.

Task 6 — Rolling estimation and parameter stability

For the rolling RV-OLS model, store the estimate of b and a HAC/Newey-West 95% confidence interval for every forecast origin. Plot the rolling coefficient and its confidence band. For rolling GARCH, store α, β, and $\alpha + \beta$; a simple parameter plot is sufficient.

Then:

(a) Compare expanding and two-year rolling forecast performance using MSE, QLIKE, correlation and average forecast level.

(b) Is the RV persistence coefficient stable? Is it statistically positive throughout most of the rolling windows?

(c) How stable is GARCH persistence? What do estimates close to one imply?

(d) If the model ranking changes when the estimation window changes, what does that suggest about the usefulness of distant historical observations?

(e) Does this prove that Bitcoin experienced a particular structural break? Explain why or why not.

Do not select the "best" rolling regression from hundreds of windows. The object being evaluated is the **forecasting rule**: re-estimate on the most recent two years, forecast one day ahead, move the window, and repeat.

Hints for Task 6 (optional)

The rolling-window loop itself — selecting the right two-year slice, re-estimating, forecasting one day ahead, and moving forward — is the core of the exercise and is left to you. The two hints below only cover statistical machinery you have not necessarily used before; they are not required, and you are welcome to derive or code your own version instead.

- **HAC/Newey-West standard errors for the RV-OLS slope.** You do not need to derive the sandwich estimator by hand. After fitting an OLS model in `statsmodels`, robust standard errors that account for heteroskedasticity and serial correlation in the residuals are available directly:

  ```
  hac_results = ols_results.get_robustcov_results(cov_type="HAC", maxlags=L)
  ```

  Choose L (e.g. 5–10 lags) and state your choice briefly in the report; an extensive search over L is not expected.

- **A confidence interval for GARCH persistence, $\alpha + \beta$.** Because $\hat{\alpha}$ and $\hat{\beta}$ are estimated jointly, they are generally correlated, so their variances cannot simply be added:

  $$\mathrm{Var}(\hat{\alpha} + \hat{\beta}) = \mathrm{Var}(\hat{\alpha}) + \mathrm{Var}(\hat{\beta}) + 2\,\mathrm{Cov}(\hat{\alpha}, \hat{\beta}).$$

  The `arch` package exposes the full parameter covariance matrix as `result.param_cov`, from which you can read off these three terms directly rather than assuming independence.

8 Part VII — Recommendation to the risk manager

Prepare a final recommendation. There is no requirement that one model dominate every metric. Your recommendation should explain:

- whether Bitcoin volatility contains a meaningful predictable component;
- whether ARCH/GARCH captures economically sensible volatility dynamics;
- whether Student-t innovations are preferable for Bitcoin;
- what the realized-variance comparison tells you about actual forecast quality;
- how the dynamics of the GARCH and RV-OLS forecasts differ, whether GARCH's greater dynamic flexibility is economically useful, and whether it actually improves OOS performance;
- how you interpret statistical significance relative to OOS performance;
- whether you prefer an expanding or rolling information set, and why;
- what these forecasts can realistically be used for in financial risk management;
- the main limitation or uncertainty you would communicate to management.

Conclude with an executive recommendation of no more than **150 words** answering:

What would you implement tomorrow, what would you use the forecast for, and how much confidence would you place in it?

Deliverable

This is a **take-home case**. Prepare a concise professional report of no more than **5 pages**, excluding the Python appendix. Your report should emphasize interpretation and decision-making rather than raw software output. Keep the code required to reproduce all results.

The case will be discussed in class. Be prepared to defend a recommendation even if another team reasonably prefers a different model. The purpose is not to discover a perfect volatility forecast; it is to understand what can be predicted, how forecast quality should be evaluated, and how much uncertainty remains.